

Ching Yao Lin

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PROFESSIONAL SUMMARY

New Grad Software Engineer and **Rice University Master of Computer Science** graduate with two software engineering internships building full-stack product tools, backend APIs, and cloud-deployed systems. Experienced across **React** and **Next.js** product interfaces, **Fastify** and **Python** API workflows, **PostgreSQL** and **MySQL** data modeling, **OAuth 2.0** and **JWT** authentication flows, and **app-controlled agentic AI workflows**.

EDUCATION

Rice University
Master of Computer Science

Houston, TX
December 2025

Feng Chia University
Bachelor of Information Engineering & Computer Science

Taichung, Taiwan
June 2024

WORK EXPERIENCE

Paycom Software

Irving, TX

Software Development Intern | React, TypeScript, Next.js, Docker, AWS Amplify

May 2025 – August 2025

- Built an **IDE-style internal prototyping platform** with React, TypeScript, Next.js, and Monaco Editor, enabling product and engineering teams to validate UI flows before production implementation.
- Developed a drag-and-drop **UI Playroom** for composing reusable UI prototypes, using Redux Toolkit to preserve component configuration and interaction state across previews.
- Containerized the application with Docker and configured **AWS Amplify CI/CD** to standardize builds, deployments, and environment setup for the internal tool.

SXB Liberal Arts & Science Tutoring Center

Yunlin, Taiwan

Software Engineer Intern | Python, SQL, Vue.js, HTML/CSS, JavaScript

May 2024 – August 2024

- Built a **MySQL-backed billing platform** with Python, SQL, and Vue.js for invoice generation, payment tracking, staff lookup, and reconciliation workflows.
- Integrated **OAuth 2.0** and **OIDC** login, JWT session handling, route-level authorization, and structured audit logging to protect staff billing workflows.
- Migrated legacy billing records with **Python ETL** scripts and normalized invoice/payment schemas, preserving historical payment lookup accuracy.

PROJECTS

GatherPoint AI

TypeScript, Next.js, Fastify, Azure OpenAI, PostgreSQL, Prisma, pgvector, Redis, Cloudflare R2

- Orchestrated an app-controlled **AI discovery workflow** that converts natural-language intent into validated planner state, SQL and pgvector retrieval, ranked event cards, and draft-event handoff.
- Built **viewer-scoped Fastify APIs** for event creation, registration, privacy-aware reads, and assistant state, backed by Clerk auth, Zod validation, **Redis rate limits**, and PostgreSQL data access.
- Shipped an **owner-scoped media lifecycle** for event images, using Cloudflare R2 presigned uploads, permission-checked reads, and background WebP conversion for secure generated and uploaded event images.

iOS Poker Session Tracking & Analytics App

Swift, SwiftUI, AWS Lambda, API Gateway, DynamoDB

- Built a **SwiftUI MVVM analytics client** for session logging, hand-history review, bankroll trends, and social activity, turning poker records into searchable mobile workflows.
- Implemented a mobile authentication layer with **RS256 JWTs** and **OAuth 2.0 PKCE** sign-in, refresh-token rotation, and JWKS distribution to secure native and API access.
- Deployed an **AWS Lambda** and **DynamoDB** backend for session, analytics, and social data, using GSIs, conditional writes, TTL cleanup, and GitHub OIDC CI/CD without long-lived AWS keys.

AI Travel Itinerary Recommendation Platform

Python, SQL, OpenAI API, Google Maps API

- Built a **retrieval-augmented recommendation pipeline** that turns user preferences, destination metadata, and time-window constraints into structured OpenAI context for constraint-aware itinerary generation.
- Implemented **SQL-backed feasibility scoring** with Google Maps API signals to rank destinations by schedule fit, travel distance, available time, and preference match, reducing impractical itinerary suggestions.

TECHNICAL SKILLS

Programming Languages: TypeScript, JavaScript, Python, SQL, Java, Swift, HTML/CSS, C/C++

Frameworks & Libraries: React, Next.js, Node.js, Fastify, Express, Redux Toolkit, Zod, Tailwind CSS, Vue.js, SwiftUI

Databases & Testing: PostgreSQL, Prisma, MySQL, DynamoDB, Redis, Playwright, Jest, Vitest

Developer Tools: Docker, CI/CD, GitHub Actions, GitLab, RESTful APIs, OpenAI API, AWS, Azure